

Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
 2. Change “Student Name” on line 3 (above) with your name.
 3. Work through the steps, **creating code and output** that fulfill each instruction.
 4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
 5. Be sure to **answer the questions** in this assignment document.
 6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
-

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the NEON_NIWO_Litter_mass_trap_Processed.csv version, again from the Processed_KEY folder).
2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
```

```
#loading packages  
library(here)
```

```
## here() starts at /home/guest/EDE_Spring2026
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2    3.5.1      v tibble     3.2.1  
## v lubridate  1.9.3      v tidyr      1.3.1  
## v purrr      1.0.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
## stamp
```

```
here()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

```
getwd()
```

```
## [1] "/home/guest/EDE_Spring2026"
```

```
#loading data
PeterPaul.chem.nutrients <- read.csv(
  here("./Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv"),
  stringsAsFactors = TRUE)

NIWOT.litter <- read.csv(here("./Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv"),
  stringsAsFactors = TRUE)

#2 #fixing dates
PeterPaul.chem.nutrients$sampldate <-
  ymd(PeterPaul.chem.nutrients$sampldate)

NIWOT.litter$collectDate <- ymd(NIWOT.litter$collectDate)
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```
#3
mytheme <- theme_minimal(base_size = 12) +
  #customizing axis labels and titles
  theme(axis.text = element_text(color = "black"),
```

```
axis.title = element_text(color = "darkblue"),
#customizing legend position
legend.position = "top",
plot.title = element_text(face = "bold", size = 14))
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (tp_ug) by phosphate (po4), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the lm method. Adjust your axes to hide extreme values (hint: change the limits using xlim() and/or ylim()).

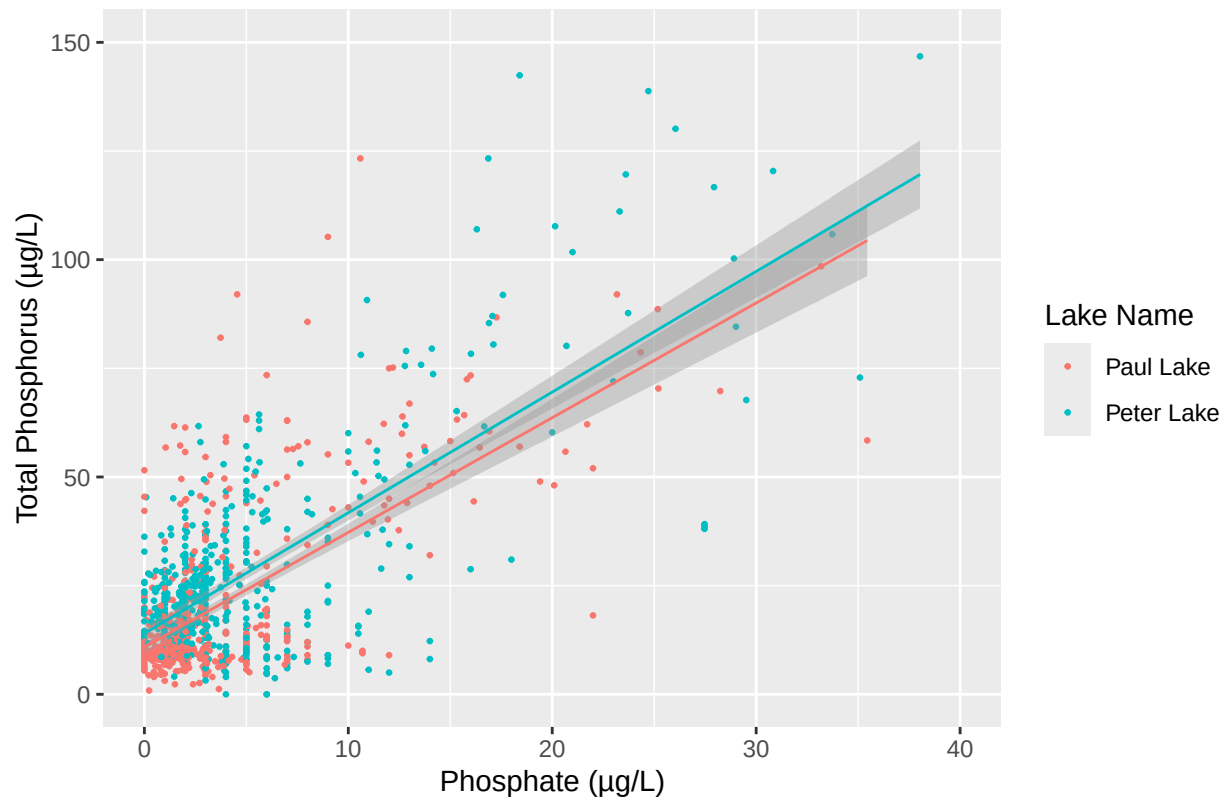
```
#4
PeterPaul.chem.nutrients %>%
  #plotting total phosphorus by phosphate
  #separate aesthetics for each lake
  ggplot(aes(x = po4, y = tp_ug, color = lakename)) +
  labs(x = "Phosphate (µg/L)",
        y = "Total Phosphorus (µg/L)",
        color = "Lake Name",
        title = "Plotting Total Phosphorus by Phosphate") +
  theme_set(mytheme) +
  geom_point(size = 0.5) +
  #adding lines of best fit
  geom_smooth(method = lm, show.legend = FALSE, linewidth = 0.5 ) +
  xlim(0,40) +
  ylim(0,150)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 21949 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 21949 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Plotting Total Phosphorus by Phosphate



#hiding extreme values

5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned. Show all months, even those where no data was collected.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis text in your theme to `element_blank()` removes the axis text (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different relative sizes when combined using `cowplot`.

```
#5 #plotting temperature
temp_plot <- PeterPaul.chem.nutrients %>%
  mutate(month_f = factor(
    month,
    level = 1:12,
    labels = month.abb)) %>%
  ggplot(aes(x = month_f,
             y = temperature_C,
             color = lakename)) +
  theme_set(mytheme) +
  theme(axis.title.x = element_blank(),
        axis.text.x = element_blank(),
        axis.title.y = element_text(size = 12)) +
```

```

geom_boxplot(size = 0.5) +
  labs(x = "Month",
       y = "Temperature",
       title = "Plotting Temperature, Total Nitrogen,
               and Total Phosphorus by Month",
       color = "Lake Name") +
  scale_x_discrete(drop=F)

#plotting total phosphorus
tp_plot <- PeterPaul.chem.nutrients %>%
  mutate(month_f = factor(
    month,
    level = 1:12,
    labels = month.abb)) %>%
  ggplot(aes(x = month_f,
             y = tp_ug,
             color = lakename)) +
  theme_set(mytheme) +
  theme(axis.title.x = element_blank(),
        axis.text.x = element_blank(),
        axis.title.y = element_text(size = 12)) +
  geom_boxplot(show.legend = FALSE,
              size = 0.5,
              outlier.size = 0.5) +
  labs(y = "Total Phosphorus (µg/L)") +
  scale_x_discrete(drop=F)

#plotting total nitrogen
tn_plot <- PeterPaul.chem.nutrients %>%
  mutate(month_f = factor(
    month,
    level = 1:12,
    labels = month.abb)) %>%
  ggplot(aes(x = month_f,
             y = tn_ug,
             color = lakename)) +
  theme_set(mytheme) +
  theme(axis.title.y = element_text(size = 12),
        axis.text.x = element_text(
          hjust = 0,
          vjust = 0.5,
          angle = -90)) +
  geom_boxplot(show.legend = FALSE,
              size = 0.5,
              outlier.size = 0.5) +
  labs(x = "Month",
       y = "Total Nitrogen (µg/L)") +
  scale_x_discrete(drop=F)

plot_grid(temp_plot, tp_plot, tn_plot, ncol = 1, align = 'v')

```

```

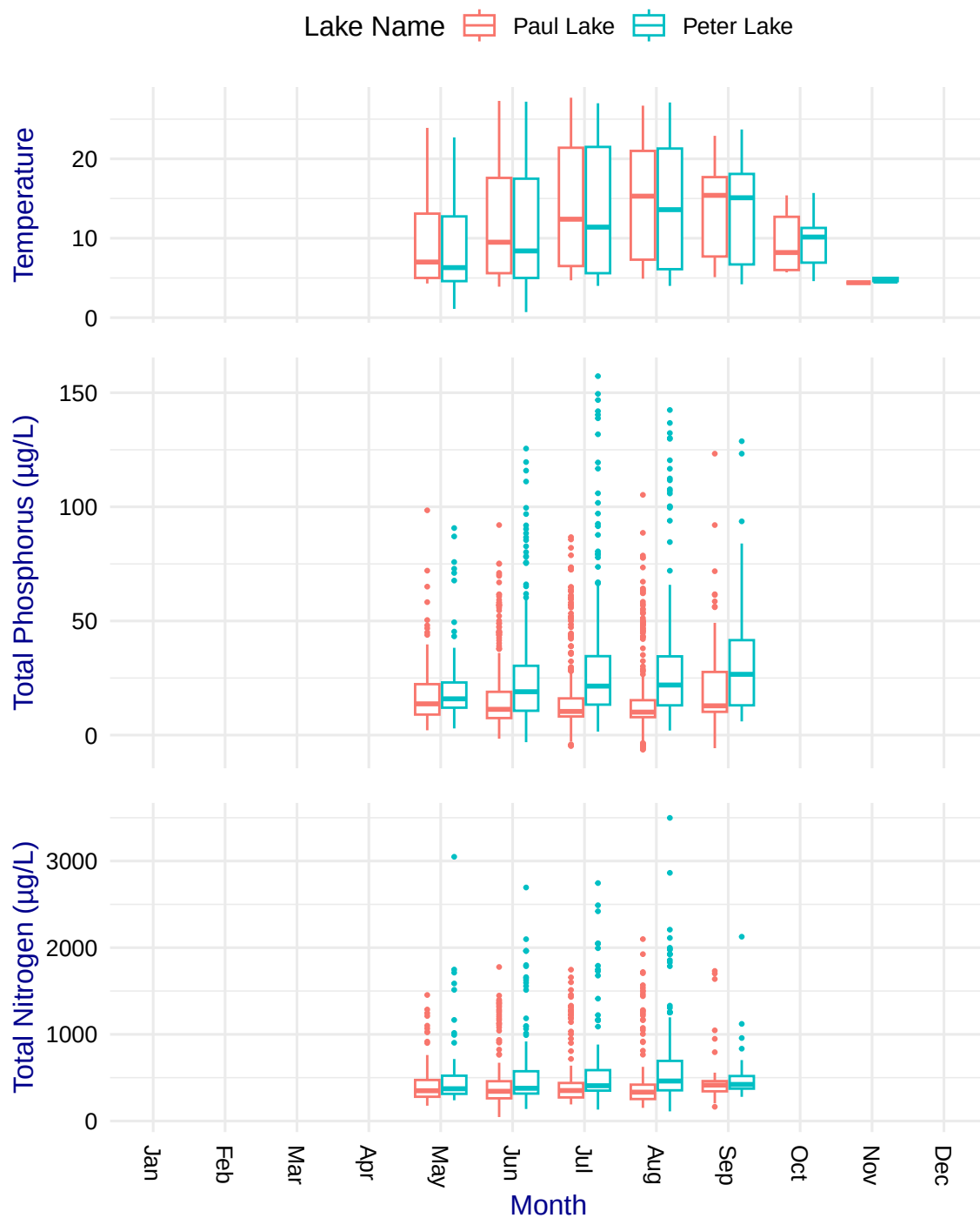
## Warning: Removed 3566 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

Plotting Temperature, Total Nitrogen, and Total Phosphorus by Month



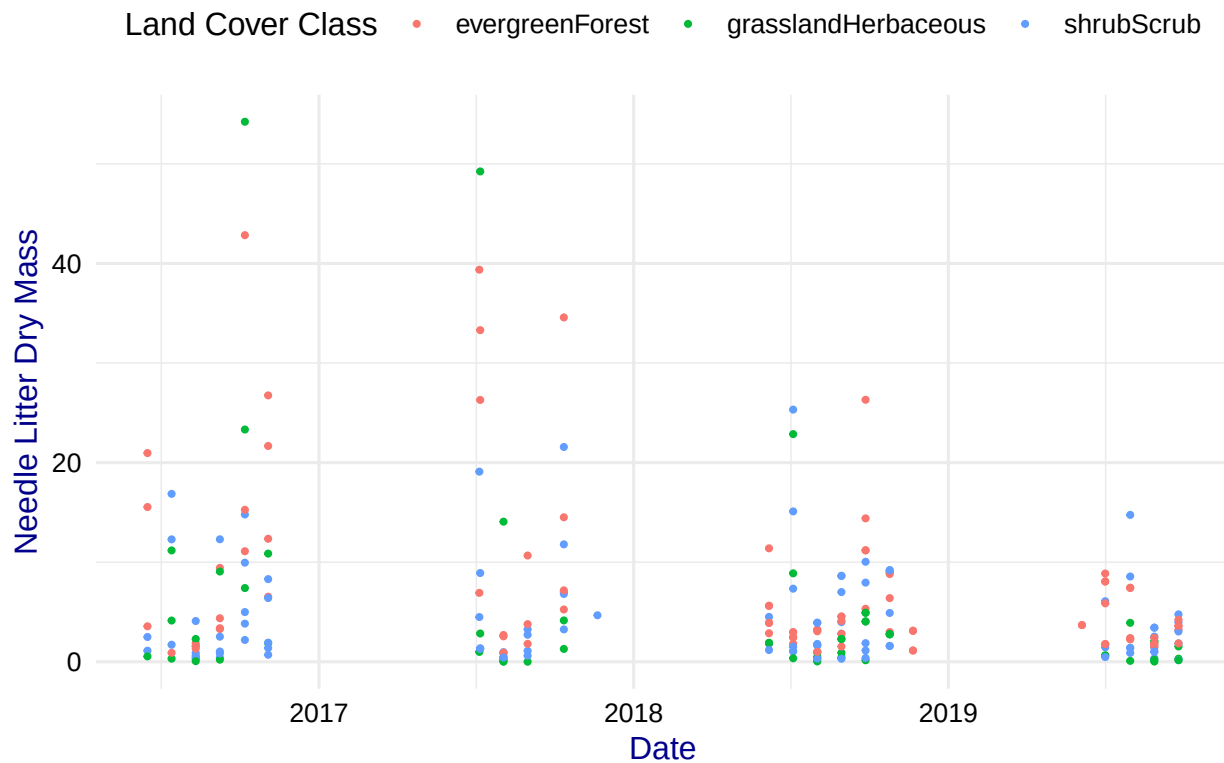
Question: What do you observe about the variables of interest over seasons and between lakes?

Answer: The mean total nitrogen and total phosphorus is generally higher in Peter Lake compared to Paul Lake. Peter Lake also has greater data variance for each of these variables. The summer months of June, July, and August have a trend of higher temperatures and total phosphorus, though I am not able to detect a seasonal trend for total nitrogen.

6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the “Needles” functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

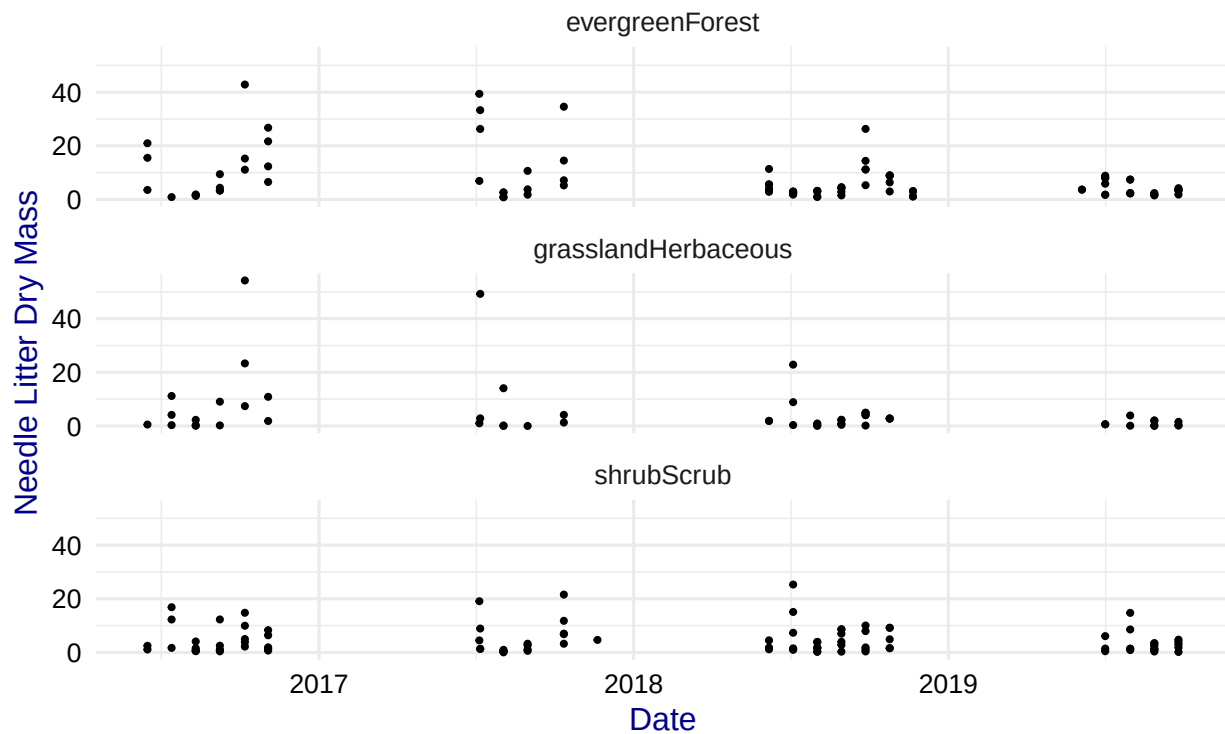
```
#6
needles_plot <-
  #subsetting needles only
  ggplot(subset(NIWOT.litter, functionalGroup == "Needles"),
    #plotting dry mass by date
    aes(x = collectDate,
        y = dryMass,
        #separating NLCD by color
        color = nlcdClass)) +
  theme_set(mytheme) +
  geom_point(size = 0.75) +
  labs(x = "Date",
    y = "Needle Litter Dry Mass",
    color = "Land Cover Class",
    title = "Plotting Needle Litter Dry Mass by Date")
print(needles_plot)
```

Plotting Needle Litter Dry Mass by Date



```
#7
needles_plot_NLCD <-
  #subsetting needles only
  ggplot(subset(NIWOT.litter, functionalGroup == "Needles"),
    #plotting dry mass by date
    aes(x = collectDate,
        y = dryMass)) +
  theme_set(mytheme) +
  geom_point(size = 0.75) +
  labs(x = "Date",
    y = "Needle Litter Dry Mass",
    title = "Plotting Needle Litter Dry Mass by Date
    and Land Cover Class") +
  #separating by facet
  facet_wrap(vars(nlcdClass), nrow=3) # facet by NLCD in three rows
  print(needles_plot_NLCD)
```


Plotting Needle Litter Dry Mass by Date and Land Cover Class



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: I find plot 6 to be more effective, because the different colors allow us to more directly visually compare the distribution of dry mass of needle litter according to the different land cover classes. Plot 7, though allowing for a more detailed view of each land cover class, almost provides too much to look at, making it hard to draw conclusions.